Exercise 01

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2      v_company_name stock.company%TYPE := 'IBM';
  3      v_current_price stock.price%TYPE;
  4  BEGIN
  5      SELECT s.price INTO v_current_price
  6      FROM stock s
  7      WHERE s.company = v_company_name;
  8
  9      DBMS_OUTPUT.PUT_LINE('Current price of ' || v_company_name ||
 10                           ' is: ' || v_current_price);
 11  END;
 12  /
Current price of IBM     is: 70

PL/SQL procedure successfully completed.
```

Exercise 02

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2      v_company_name stock.company%TYPE := 'IBM';
  3      v_current_price stock.price%TYPE;
  4  BEGIN
  5      SELECT s.price INTO v_current_price
  6      FROM stock s
  7      WHERE s.company = v_company_name;
  8
  9      DBMS_OUTPUT.PUT_LINE('Company: ' || v_company_name);
 10      DBMS_OUTPUT.PUT_LINE('Current Price: ' || v_current_price);
 11      DBMS_OUTPUT.PUT_LINE('Message: ');
 12
 13      IF v_current_price < 45 THEN
 14          DBMS_OUTPUT.PUT_LINE('Current price is very low');
 15      ELSIF v_current_price < 55 THEN
 16          DBMS_OUTPUT.PUT_LINE('Current price is low');
 17      ELSIF v_current_price < 65 THEN
 18          DBMS_OUTPUT.PUT_LINE('Current price is medium');
 19      ELSIF v_current_price < 75 THEN
 20          DBMS_OUTPUT.PUT_LINE('Current price is medium high');
 21      ELSE
 22          DBMS_OUTPUT.PUT_LINE('Current price is high');
 23      END IF;
 24  END;
 25  /
Company: IBM
Current Price: 70
Message:
Current price is medium high

PL/SQL procedure successfully completed.
```

Exercise 03

SIMPLE LOOP

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2      v_row NUMBER := 9;
  3      v_col NUMBER;
  4      v_output VARCHAR2(200);
  5  BEGIN
  6      LOOP
  7          EXIT WHEN v_row < 1;
  8          v_col := 1;
  9          v_output := '';
 10
 11          LOOP
 12              EXIT WHEN v_col > v_row;
 13              v_output := v_output || v_row;
 14              v_col := v_col + 1;
 15          END LOOP;
 16
 17          DBMS_OUTPUT.PUT_LINE(v_output);
 18          v_row := v_row - 1;
 19      END LOOP;
 20  END;
 21  /
999999999
88888888
7777777
666666
55555
4444
333
22
1

PL/SQL procedure successfully completed.
```

WHILE LOOP

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2      v_row NUMBER := 9;
  3      v_col NUMBER;
  4      v_output VARCHAR2(200);
  5  BEGIN
  6      WHILE v_row >= 1 LOOP
  7          v_col := 1;
  8          v_output := '';
  9
 10          WHILE v_col <= v_row LOOP
 11              v_output := v_output || v_row;
 12              v_col := v_col + 1;
 13          END LOOP;
 14
 15          DBMS_OUTPUT.PUT_LINE(v_output);
 16          v_row := v_row - 1;
 17      END LOOP;
 18  END;
 19  /
999999999
88888888
7777777
666666
55555
4444
333
22
1

PL/SQL procedure successfully completed.
```

FOR LOOP

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2      v_output VARCHAR2(200);
  3  BEGIN
  4      FOR v_row IN REVERSE 1..9 LOOP
  5          v_output := '';
  6
  7          FOR v_col IN 1..v_row LOOP
  8              v_output := v_output || v_row;
  9          END LOOP;
 10
 11          DBMS_OUTPUT.PUT_LINE(v_output);
 12      END LOOP;
 13  END;
 14  /
999999999
88888888
7777777
666666
55555
4444
333
22
1

PL/SQL procedure successfully completed.
```

Exercise 04

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2      v_bonus_qty NUMBER;
  3  BEGIN
  4      FOR purchase_rec IN (SELECT * FROM purchase) LOOP
  5          -- Determine bonus quantity based on purchase date
  6          IF purchase_rec.pdate < TO_DATE('01-JAN-2000', 'DD-MON-YYYY') THEN
  7              v_bonus_qty := 150;
  8          ELSIF purchase_rec.pdate < TO_DATE('01-JAN-2001', 'DD-MON-YYYY') THEN
  9              v_bonus_qty := 100;
 10          ELSIF purchase_rec.pdate < TO_DATE('01-JAN-2002', 'DD-MON-YYYY') THEN
 11              v_bonus_qty := 50;
 12          ELSE
 13              v_bonus_qty := 0; -- No bonus for purchases after 2002
 14          END IF;
 15
 16          -- Update the quantity if bonus applies
 17          IF v_bonus_qty > 0 THEN
 18              UPDATE purchase p
 19              SET p.qty = p.qty + v_bonus_qty
 20              WHERE p.clno = purchase_rec.clno
 21                AND p.stockno = purchase_rec.stockno
 22                AND p.pdate = purchase_rec.pdate;
 23
 24              DBMS_OUTPUT.PUT_LINE('Updated purchase for client ' ||
 25                                   purchase_rec.clno || ' with bonus: ' || v_bonus_qty);
 26          END IF;
 27      END LOOP;
 28
 29      COMMIT;
 30      DBMS_OUTPUT.PUT_LINE('All updates completed successfully.');
 31  END;
 32  /
             AND p.stockno = purchase_rec.stockno
                                           *
```

Exercise 05

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2      CURSOR purchase_cur IS
  3          SELECT * FROM purchase;
  4
  5      purchase_rec purchase%ROWTYPE;
  6      v_bonus_qty NUMBER;
  7  BEGIN
  8      -- Open the cursor
  9      OPEN purchase_cur;
 10
 11      -- Fetch first row
 12      FETCH purchase_cur INTO purchase_rec;
 13
 14      -- Loop while rows exist
 15      WHILE purchase_cur%FOUND LOOP
 16          -- Determine bonus quantity based on purchase date
 17          IF purchase_rec.pdate < TO_DATE('01-JAN-2000', 'DD-MON-YYYY') THEN
 18              v_bonus_qty := 150;
 19          ELSIF purchase_rec.pdate < TO_DATE('01-JAN-2001', 'DD-MON-YYYY') THEN
 20              v_bonus_qty := 100;
 21          ELSIF purchase_rec.pdate < TO_DATE('01-JAN-2002', 'DD-MON-YYYY') THEN
 22              v_bonus_qty := 50;
 23          ELSE
 24              v_bonus_qty := 0;
 25          END IF;
 26
 27          -- Update if bonus applies
 28          IF v_bonus_qty > 0 THEN
 29              UPDATE purchase p
 30              SET p.qty = p.qty + v_bonus_qty
 31              WHERE p.clno = purchase_rec.clno
 32                AND p.stockno = purchase_rec.stockno
 33                AND p.pdate = purchase_rec.pdate;
 34
 35              DBMS_OUTPUT.PUT_LINE('Updated purchase for client ' ||
 36                                   purchase_rec.clno || ' with bonus: ' || v_bonus_qty);
 37          END IF;
 38
 39          -- Fetch next row
 40          FETCH purchase_cur INTO purchase_rec;
 41      END LOOP;
 42
 43      -- Close the cursor
 44      CLOSE purchase_cur;
 45
 46      COMMIT;
 47      DBMS_OUTPUT.PUT_LINE('All updates completed successfully.');
 48  EXCEPTION
 49      WHEN OTHERS THEN
 50          -- Handle any errors and close cursor if open
 51          IF purchase_cur%ISOPEN THEN
 52              CLOSE purchase_cur;
 53          END IF;
 54          DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);
 55          ROLLBACK;
 56  END;
 57  /
              AND p.stockno = purchase_rec.stockno
                                               *
```